morphagene

_travelers guide _for navigating wormholes

2.9 min per reel	300 splices per reel	32 reels per card

[rec] press [rec] hold 3 s = keep holding › then 12:00 = knob straight up ◆ = options.txt only

s.o.s. = sound on sound · tla = time lag accumulation · eosg = end of splice / gene

max reel length = 2.9_min | 174_sec

yellow = hold

01 buttons mg_01

[rec]	record into this splice · press again to stop. empty reel = new splice
[rec] + [splice]	**record into a new splice** at the end of the reel › [rec] stops
[splice]	drop a splice marker at the playhead (works while recording)
[shift]	next splice · also mounts the microsd card
release	single presses act on release (except rec stop), so combos never misfire
[rec] LED	lit while recording · strobes while it waits for a clock
[rec] ◆ rsop 1	swapped: a plain press starts a new splice, [rec] + [splice] records into this one

02 vari-speed mg_02

turn [vari-speed]	12:00 = stop · cw = forward · ccw = reverse, faster the further you go
1/1 green	original speed at ~2:30 forward / ~9:30 reverse
[vari-speed] cv	±4 v through the attenuverter
playback only	recording is never sped up: set it back to green to hear it as captured
[vari-speed] cv ◆ vsop 1 / 2	**1** = 1v/oct both ways · **2** = forward-only 1v/oct, stop at full ccw

__fig.01 : vari-speed

activity window colors



range about +12 / -26 semitones · finest near 12:00 · window colors

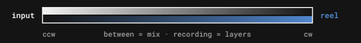
03 delete mg_03

[shift] + [splice]	**delete marker**: this splice merges with the next
[shift] + [splice] 3 s	**delete all markers**: one big splice, audio kept
[shift] + [rec]	**delete this splice's audio**
[shift] + [rec] 3 s	**clear the reel**: every marker and all audio
splice window	flashes to confirm any erase · wait for [shift] to stop flashing

* no deleting with a clock patched to play.

04 sound on sound mg_04

turn [s.o.s.]	ccw = live input · cw = reel only · between = mix
[rec] on a loop	**time lag accumulation**: records over the splice as it loops, s.o.s. sets how much old audio survives
[s.o.s.] cv	0–8 v, normalled to +8 v · knob turns into its attenuator
no input	acts as a vca for the loop, or a crossfader between live and loop
tla drift	every pass is re-pitched: off 1/1 (or morph past 2:30) walks out of range
[rec] on a loop ◆ inop 1	records the input only: tla becomes a single-repeat delay



05 organize · reels mg_05

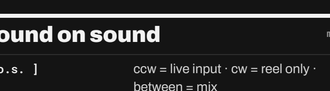
turn [organize]	pick the next splice: evenly spread, lands when this splice / gene ends
[splice] + [rec]	**reel mode** in / out (card mounted, reel window flashes)
in reel mode	[organize] picks a reel · s.o.s. cw = preview · [splice] + [rec] loads · [shift] + [rec] deletes it
[organize] / [shift] ◆ omod 1	jump right away instead of waiting (may click)



mg1 … mg9 · mga … mgw.wav last slot flashes pink / white = new reel

06 microsound mg_06

turn [gene-size]	ccw = whole splice · cw = tiny genes (grains). never destructive
turn [slide]	travel through the splice: scrubs and moves the play start point
turn [morph]	gaps › seamless › overlaps › stacked genes with pitch & pan
cv	gene-size 0–8 v, attenuator · slide 0–8 v, attenuverter · morph 0–5 v
genes ◆ gnsm 1	stronger windowing, fades at small gene sizes



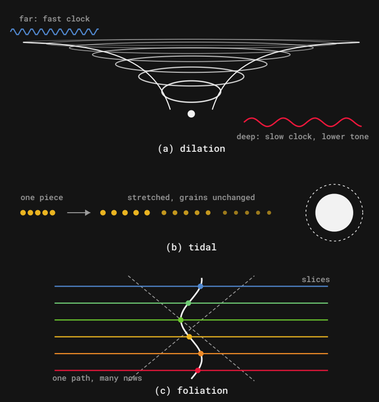
07 input gain mg_07 · mg203+

[rec] + [shift]	step through four input gain settings:
ccw [s.o.s.]	hear the input only while you set it · inputs are ac coupled, line to modular



__fig.02 : the well

mm_03



08 clock mg_08

[clk] + morph ccw of 11:00	**gene shift** (red): every clock jumps to the next gene
[clk] + morph 11:00 +	**time stretch** (blue): clock sets the pace, vari-speed is pitch only
needs	gene-size below a full splice · a clock at least every 3.5 s · tops out ~18 hz
[clk] + [rec]	rec strobes, then starts and stops on the next clock
[clk] ◆ ckop 1 / 2	**1** = gene shift only · **2** = time stretch only



09 gates · cv mg_09

[play]	high = loop · low = stop at the end · rising edge = restart from slide. unpatched = loops
[rec] [splice] [shift]	gate inputs: remote buttons (2.5 v +)
[eosg]	gate at the end of every splice / gene: fires faster with gene-size & morph
[cv out]	envelope follower, 0–8 v
[play] ◆ pmod 1 / 2	**1** = momentary (plays while high) · **2** = trigger loop
[cv out] ◆ cvop 1	a ramp timed to the gene
[in] r ◆ pmin 1	phase-modulates the reel when the left input is empty

10 microsd mg_10

[shift] LED	on = card mounted · flashing = busy, don't pull the card
copy a reel	load it › pull the card › wait for shift to go dark › reinsert › [shift]
options.txt	made on mount · edit it, reinsert, [shift]
power	20 hp · +12 v 165 ma · −12 v 20 ma

__fig.03 : front panel

01–10 = sections · 12–22 = hardware

[01] __rec : record · tla · stop
[02] __vari-speed : speed & direction
[03] __splice : marker · eos light
[04] __s.o.s. : input / reel blend
[05] __organize : next splice · reel
[06] __gene-size : grain length
[07] __in : audio in l (mono) / r
[08] __clk : shift · stretch · rec sync
[09] __play : gate · retrigger
[10] __microsd : reels on fat32

[12] __shift : next splice · mount
[13] __out : audio out l (mono) / r
[14] __morph : gene overlap
[15] __slide : travel · start point
[16] __windows : speed · morph colors
[17] __reel : reel color · clock
[18] __eosg : end of splice / gene
[19] __cv out : envelope · window
[20] __cv ins : every knob has one
[21] __splice window : splice color
[22] __gate ins : rec · splice · shift



_where things hide

11 · mg_11

options.txt

card root · edit the number, re-mount

- **vsop** 0 classic · 1 1v/oct · 2 forward-only 1v/oct
- **inop** 0 record the s.o.s. mix · 1 input only
- **pmin** 1 = right input phase-mods the reel
- **omod** 0 wait for the end · 1 jump now
- **gnsm** 0 classic · 1 smooth gene windows
- **rsop** 1 = swap rec / rec + splice
- **pmod** 0 classic · 1 momentary · 2 trigger loop
- **ckop** 0 hybrid · 1 gene shift · 2 time stretch
- **cvop** 0 envelope · 1 gene ramp
- **mcr1–3** mg203+: chord ratios for genes 2–4 at high morph · 0.0625–16 · minus = reverse · 1 = unison

microsd card

- **format** fat32 · up to 32 reels in the root
- **reels** .wav, 48 khz 32-bit float stereo · 2.9 min max
- **splices** up to 300 per reel · markers live inside the .wav
- **no card** one reel only, no reel mode
- **morpmarker** m4l device for ableton live: writes splice markers into a reel

home position

plain 1/1 loop playback



window colors

reels & splices cycle in this order, then repeat

reel · splice	1 2 3 4 5 6 7 8	morph	gaps · overlaps · seamless · time stretch · gene shift
vari-speed	stop · 1/1 · +1 oct · −1 oct	reel	pink / white = new reel · flashing = reel mode · clock in
gain	−3 db · modular · +6 db · +12 db	LEDs	shift on = mounted · flashing = busy · splice = end of splice / gene